13. Sodium nitroprusside Test

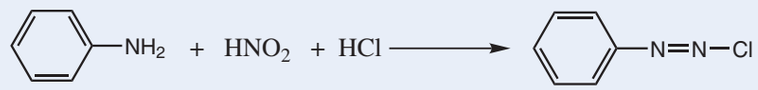
The anion of the ketone formed by a alkali reacts with nitroprusside ion to form a red coloured complex.this test is not given by aldehydes.

$$CH_3COCH_3 \xrightarrow{^-OH} CH_3COCH_2^- + H_2O$$

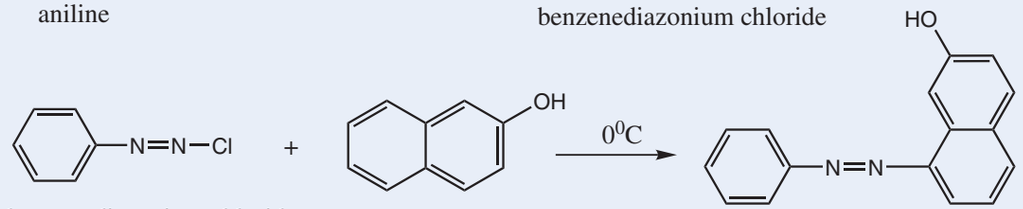
$$\underset{\text{sodium nitro prusside}}{[Fe(CN)_5\,NO]^{2-}} + CH_3COCH_2^- \longrightarrow \underset{\text{(Red coloured complex)}}{[Fe(CN)_5\,NO.CH_3COCH_2]^{3-}}$$

14. Azo-Dye Test

This test is given by aromatic primary amines. Aromatic primary amines react with nitrous acid to form diazonium salts. These diazonium salts undergo coupling reaction with β-naphthol to form orange coloured azo dye.



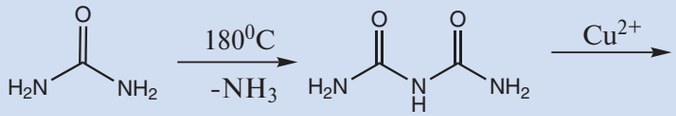
aniline — benzenediazonium chloride



benzenediazonium chloride + β-naphthol → Azo dye (Orange red)

15. Biuret test

On strong heating diamide (like urea) forms biuret, which forms a copper complex with Cu^{2+} ions from copper sulphate solution. This copper –biuret complex is deep violet coloured.



urea (Excess) → biuret → $[Cu(Biuret)_4]^{2+}$ complex (violet colour)

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